Customer Churn Feature Analysis

Business Understanding

- A telecommunications company has been losing its money due to customer churn. The company wants to know what is contributing to the loss and curb it.

Objective

- This project aims to determine whether a customer will soon churn

Syriatel telecom services based on the significance of usage features.

Key Questions

1. Which feature contributes most to customer churn?

2. Which features are the top 3 contributors of customers churn?

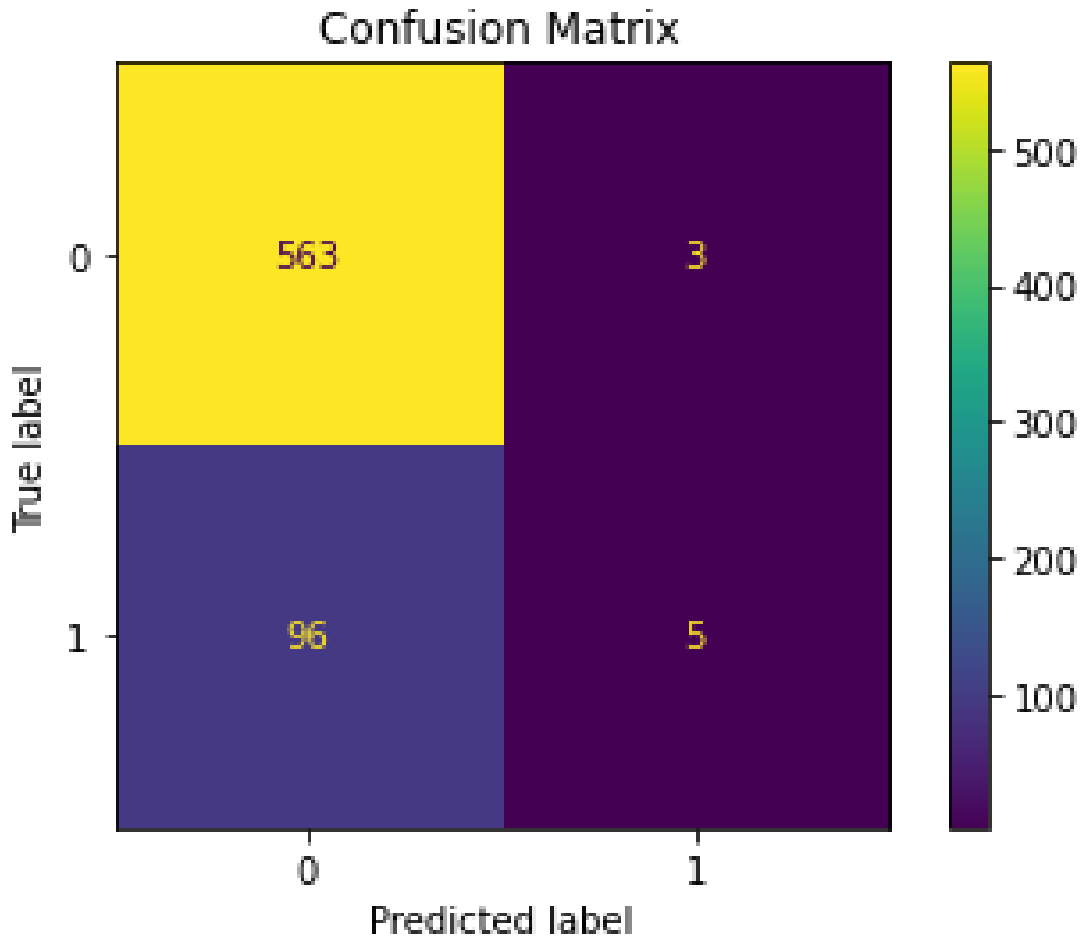
3. Which feature least contributes to customer churn?

Data Understanding

- The data used was obtained from Syriatel Telecommunications company and contained 3333 rows and 21 columns.
- The columns included:

state, account length, area code, phone number, international plan, voice mail plan, number vmail messages, total day minutes, total day calls, total day charge , total eve minutes, total eve calls, total eve charge , total night minutes, total night calls, total night charge , total intl minutes, total intl calls, total intl charge , customer service calls, churn

Confusion matrix to show the predictive ability of the model.

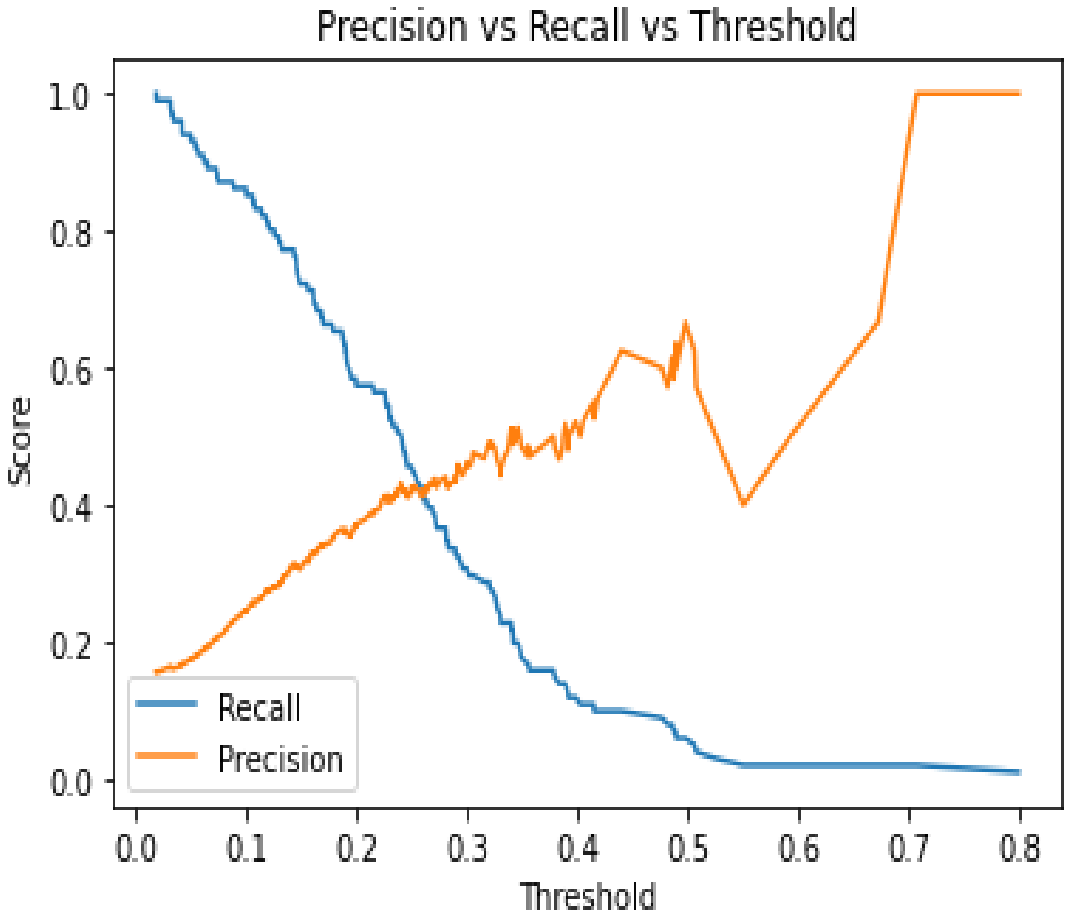


TP = 5, TN =563 , FP=3 , FN= 96. This confusion matrix shows that the model is able to predict negative non-churners more than non-churners.

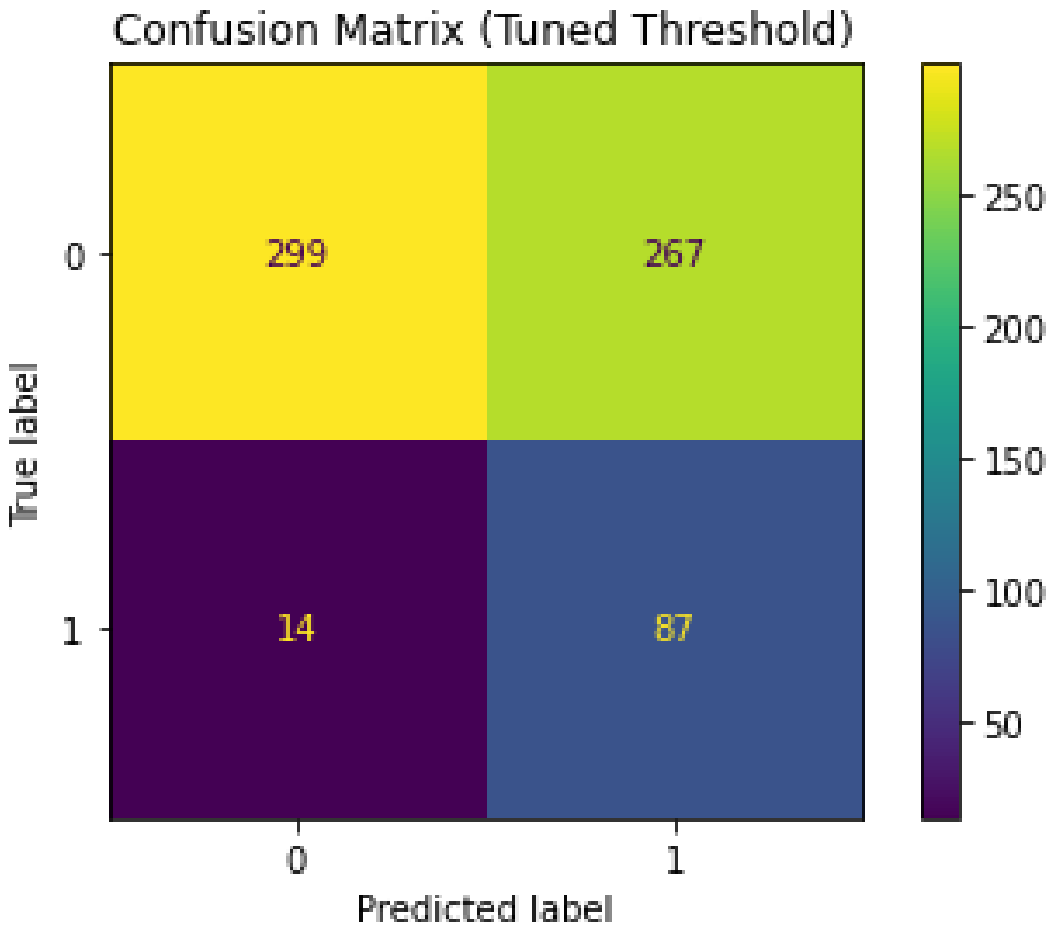
- This model had a recall score of 0.99 for non-churners and 0.05 for churners.

This meant was not able to predict churners well.

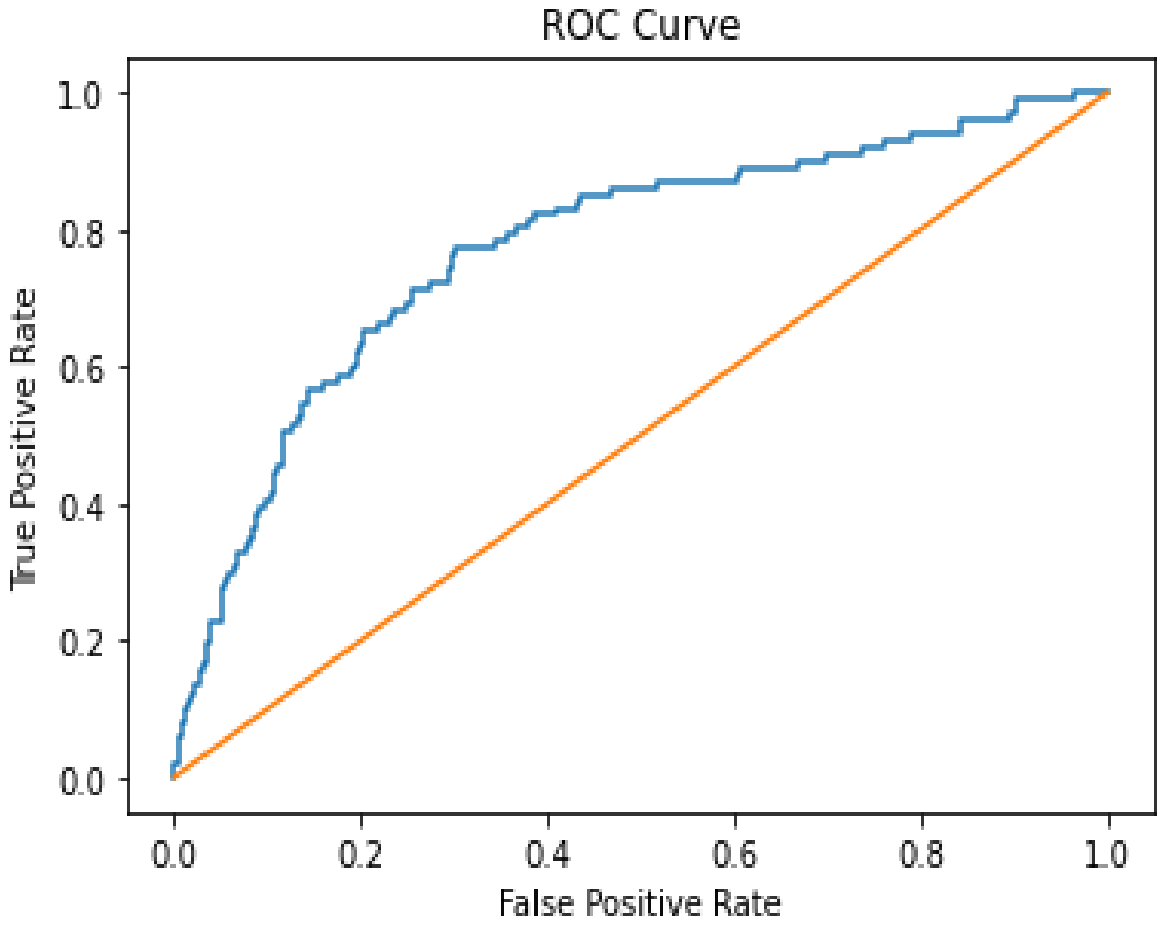
This visualization was used to choose a threshold that would have a higher recall score



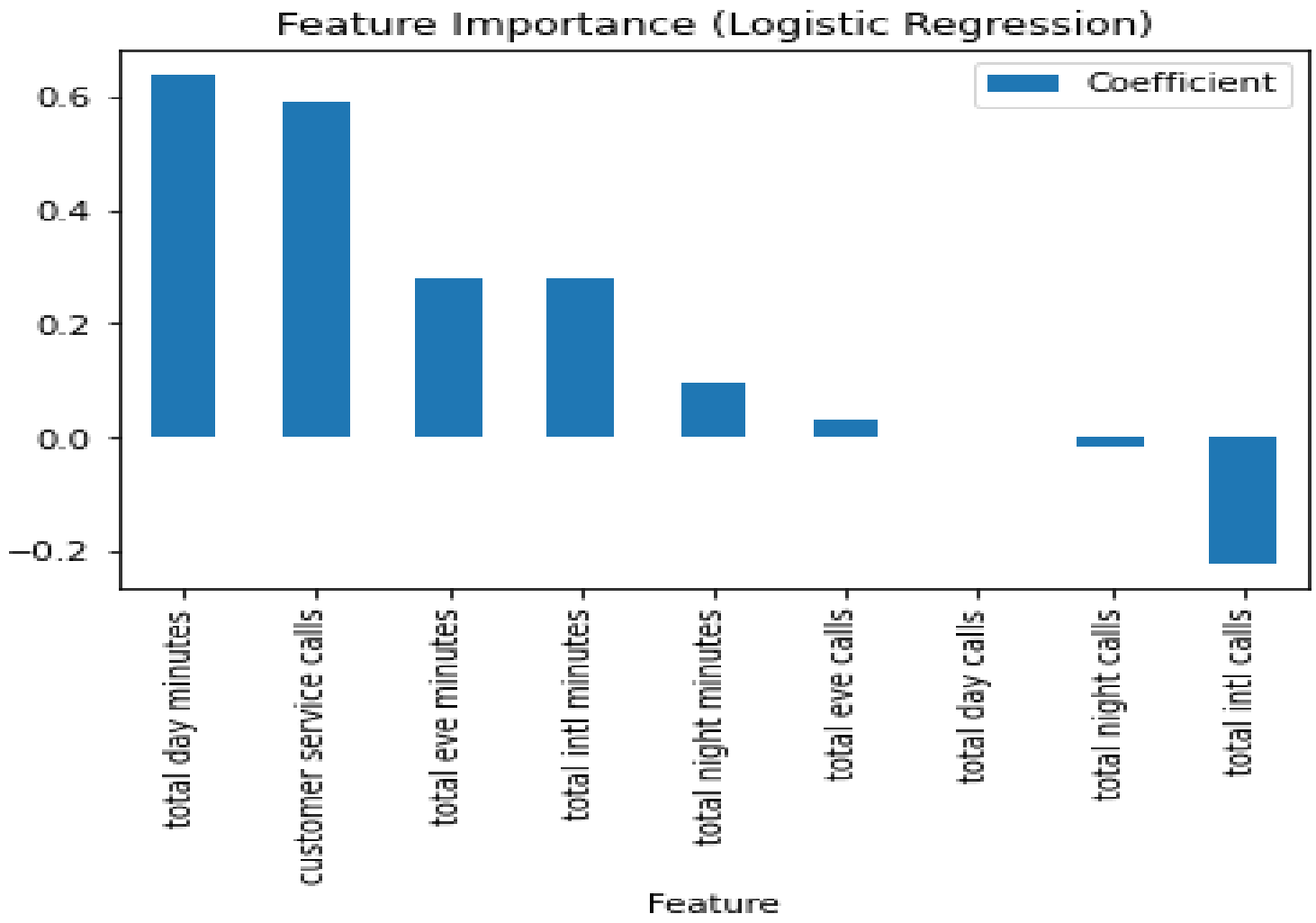
After tuning the threshold, the model was able to predict churners better



This ROC curve shows that the overall performance of the model is good as it lies to the left to a great extent



This shows how each of the features contributed to customer churn.



Key Findings

- Total day minutes is the feature that drives customer churn the most
- Total intl calls were the least cause of customer churn.
- Having an international plan was a key determinant in determining customer churn.
- Total intl minutes and total eve minutes had the same significance in causing customer churn.

Recommendations

1. Come up with loyalty incentives for high usage customers and usage based pricing plans to make their services cost effective to high users.

2. Ensure they have a reasonable cost-benefit international plan.

3. Monitor churn trends and adjust pricing or features to address upcoming customer pain points